

Quantum Foundations Seminar – Syllabus

Dr. Beate Asenbeck and Dr. Viktoria Kabel

Week 1: Introduction

Introductory lecture and group assignment.

Week 2: EPR and Bell

Main Reading:

- Einstein, Podolsky, Rosen (1935): Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?

Additional Reading:

- Bell (1964): On the Einstein Podolsky Rosen Paradox
- Clauser, Horne, Shimony, Holt (1969): Proposed Experiment to Test Local Hidden-Variable Theories

Week 3: de Broglie and Bohm

Main Reading:

- Bohm (1957): A Suggested Interpretation of the Quantum Theory in Terms of “Hidden” Variables. I

Additional Reading:

- Bohm (1957): A Suggested Interpretation of the Quantum Theory in Terms of “Hidden” Variables. II
- Tumulka (2021): Bohmian Mechanics (in *The Routledge Companion to Philosophy of Physics*)

Week 4: Contextuality I (definitions)

Main Reading:

- Mermin (1993): Hidden variables and the two theorems of John Bell

Additional Reading:

- Spekkens (2005): Contextuality for preparations, transformations, and unsharp measurements
- Klyachko, Can, Binicioğlu, Shumovsky (2008): Simple Test for Hidden Variables in Spin-1 Systems

Week 5: Contextuality II (applications)

Main Reading:

- Kleinmann (2011): Memory cost of quantum contextuality

Additional Reading:

- Howard, Wallman, Veitch, Emerson (2014): Contextuality supplies the ‘magic’ for quantum computation
- Horodecki⁴, Pawłowski, Bourennane (2010): Contextuality offers device-independent security

Week 6: Quantum-to-Classical Transition I (decoherence)

Main Reading:

- Zurek (1991): Decoherence and the Transition from Quantum to Classical

Additional Reading:

- Zeh (1969): On the Interpretation of Measurement in Quantum Theory
- Albrecht (1992): Investigating decoherence in a simple system

Week 7: Quantum-to-Classical Transition II (coarse-graining)

Main Reading:

- Kofler, Brukner (2007): Classical World Arising out of Quantum Physics under the Restriction of Coarse-Grained Measurements

Additional Reading:

- Bibak, Cepollaro, Sánchez, Dakić, Brukner (2025): The classical limit of quantum mechanics through coarse-grained measurements

Week 8: Many-Worlds Interpretation

Main Reading:

- Everett (1957): “Relative State” Formulation of Quantum Mechanics

Additional Reading:

- Wallace (2010): The Everett Interpretation

Week 9: Copenhagen Interpretation

Main Reading:

- Bohr (1935): Can Quantum-Mechanical Description of Physical Reality be Considered Complete?

Additional Reading:

- Hermann (1935): The Foundations of Quantum Mechanics in the Philosophy of Nature
- Brukner (2017): On the Quantum Measurement Problem

Week 10: Wigner's Friend**Main Reading:**

- Frauchiger, Renner (2018): Quantum Theory cannot consistently describe the use of itself

Additional Reading:

- Wigner (1967): Remarks on the Mind-Body Question
- Bong et al. (2020): A strong no-go theorem on the Wigner's friend paradox

Week 11: Relative Facts (QBism and Relational QM)**Main Reading:**

- Fuchs, Mermin, Schack (2014): An introduction to QBism with an application to the locality of quantum mechanics

Additional Reading:

- Rovelli (2022): The Relational Interpretation (in *The Oxford Handbook of the History of Quantum Interpretations*)
- Pienaar (2021): QBism and Relational Quantum Mechanics compared

Week 12: Reconstruction of QM**Main Reading:**

- Popescu, Rohrlich (1998): Causality and Nonlocality as Axioms for Quantum Mechanics

Additional Reading:

- Hardy (2002): Why Quantum Theory?
- Brukner, Zeilinger (2009): Information Invariance and Quantum Probabilities

Week 13: Quantum Foundations and Gravity**Main Reading:**

- Marletto, Vedral (2017): Gravitationally Induced Entanglement between Two Massive Particles is Sufficient Evidence of Quantum Effects in Gravity

Additional Reading:

- Huggett, Linnemann, Schneider (2023): Quantum Gravity in a Laboratory?

Week 14: Foundational Tests in Experiment

Main Reading:

- Giustina et al. (2015): Significant-Loophole-Free Test of Bell's Theorem with Entangled Photons

Additional Reading:

- Leupold et al. (2018): Sustained State-Independent Quantum Contextual Correlations from a Single Ion
- Pedalino et al. (2025): Probing quantum mechanics with nanoparticle matter-wave interferometry